

# run\_sjsdm\_selected\_candidate\_folds.R

September 10, 2026

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| run_sjsdm_selected_candidate_folds               |
| <i>Run Selected sjSDM Candidate Across Folds</i> |

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## Description

Refits one selected regularization candidate independently in every fold and materializes aligned out-of-fold probabilities.

## Usage

```
run_sjsdm_selected_candidate_folds(  
  data_assignments = NULL,  
  data_selected_candidate = NULL,  
  data_sample_ids = NULL,  
  taxon_names = NULL,  
  prepare_fold_function = NULL,  
  fit_function = NULL,  
  predict_function = NULL,  
  seed = 900723L  
)
```

## Arguments

- data\_assignments**  
Location-level assignment table accepted by [run\_sjsdm\_tuning\_candidates()].
- data\_selected\_candidate**  
One-row selected candidate table containing the candidate identifier, the six sjSDM regularization parameters, and regularization source.
- data\_sample\_ids**  
Sample metadata with 'dataset\_name' and 'age'. Optional sample, row, and location identifiers are derived when absent.

|                                    |                                                                                                                                                                                                                                                                      |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>taxon_names</code>           | Unique character vector defining the full response-taxon order.                                                                                                                                                                                                      |
| <code>prepare_fold_function</code> | Injectable function called with training and test row indices plus repeat and fold identifiers. It must return train/test inputs, retained train/test response matrices, the full test response matrix, aligned test sample identifiers, and the fold taxon mapping. |
| <code>fit_function</code>          | Injectable function called with the prepared training input, selected candidate, and deterministic fit seed.                                                                                                                                                         |
| <code>predict_function</code>      | Injectable function called with the fitted object and prepared test input. It must return probabilities aligned to retained test observations.                                                                                                                       |
| <code>seed</code>                  | Single non-negative integer used to derive stable fold fit seeds.                                                                                                                                                                                                    |

### Details

The prediction table deliberately omits the candidate identifier. Selected candidate provenance remains in the separate fold diagnostics and selected candidate artifact. Taxa not retained in a training fold remain as rows with missing probabilities and their taxon-mapping status.

### Value

Named list with data-prediction and data-diagnostic tibbles. Predictions contain one row per repeat, sample, and full-response taxon, including observed values, predictions, training-fold null probabilities, and explicit status. Diagnostics record one structured fit status per fold.

### Examples

```
## Not run:
run_sjsdm_selected_candidate_folds(
  data_assignments = data_assignments,
  data_selected_candidate = data_selected_candidate,
  data_sample_ids = data_sample_ids,
  taxon_names = taxon_names,
  prepare_fold_function = prepare_fold_function,
  fit_function = fit_function,
  predict_function = predict_function
)

## End(Not run)
```

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